

OLTP Sales Management System

1. Introduction

The **OLTP Sales Management System** is a computer-based application designed to manage and record sales transactions in real time. Businesses use such systems to store and process operational data such as customer details, product information, store locations, and sales transactions.

OLTP stands for **Online Transaction Processing**. It refers to systems that handle a large number of small transactions quickly and efficiently. Examples include supermarket billing systems, banking transactions, and online shopping systems.

This project demonstrates how an OLTP system works by using a **Sales Management System** that records and manages sales operations using a structured database and a web application built with **Spring Boot**.

The system allows businesses to:

- Manage customer information
- Track products and inventory
- Store location details
- Record and process sales transactions

All these operations are performed in **real time**, ensuring that the database remains accurate and up to date.

2. System Architecture

The system is developed using **Spring Boot**, a Java-based framework used to build web applications. The architecture follows a layered structure that separates different parts of the system to make it easier to manage and maintain.

The architecture consists of the following layers:

2.1 User Interface (Web Browser)

This is the part where users interact with the system through a web browser. Users can view data, enter new records, and manage sales information.

2.2 Controllers

Controllers handle incoming requests from users. When a user performs an action (such as adding a customer or viewing products), the controller receives the request and sends it to the appropriate service.

2.3 Services

The service layer contains the **business logic** of the system. It processes data and performs operations such as calculating totals, checking stock availability, or validating transactions.

2.4 Repositories

Repositories are responsible for communicating with the database. They perform operations such as saving, updating, deleting, and retrieving records from the database.

2.5 Database

The system uses an **H2 database**, which is a lightweight in-memory database used for development and testing purposes. It stores all the data related to customers, products, locations, and sales.

Architecture Flow

User (Web Browser)

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Controllers

↓

Services

↓

Repositories

↓

Database

3. Database Design

The system database contains **four main tables** that store all the necessary information for managing sales operations.

These tables are:

1. Customers
2. Products

3. Locations

4. Sales

Each table stores specific types of data and is connected with other tables through relationships.

4. Customers Table

The **Customers table** stores information about the people who purchase products from the business.

Main Attributes

- Customer ID – Unique identifier for each customer
- Name – Customer's full name
- Email – Customer's email address
- Phone Number – Contact number
- Address – Customer's residential address
- Status – Indicates whether the customer is active, inactive, or suspended
- Created Date – Date when the customer was added to the system
- Updated Date – Date when customer information was last updated

Purpose

This table allows the system to track customer information and maintain records of customer purchases.

5. Products Table

The **Products table** stores details about the products that the business sells.

Main Attributes

- Product ID – Unique identifier for each product
- Product Name – Name of the product
- SKU – Stock Keeping Unit (unique product code)

- Category – Type of product (Electronics, Clothing, etc.)
- Price – Selling price of the product
- Stock Quantity – Number of items currently available
- Status – Indicates whether the product is available, out of stock, or discontinued

Purpose

This table helps the system manage inventory and track product availability.

6. Locations Table

The **Locations table** stores information about business locations such as stores or warehouses.

Main Attributes

- Location ID – Unique identifier for each location
- Location Name – Name of the store or warehouse
- Location Code – Unique code for each location
- Type – Type of location (Retail store, Warehouse, Online store)
- Address – Physical location address
- Manager Name – Name of the location manager
- Operating Hours – Business operating hours

Purpose

This table helps track where sales transactions occur and allows businesses to manage multiple locations.

7. Sales Table

The **Sales table** records every transaction made in the system. It is the central table of the database.

Main Attributes

- Sale ID – Unique identifier for each transaction

- Customer ID – Reference to the customer who made the purchase
- Product ID – Reference to the product sold
- Location ID – Reference to the location where the sale occurred
- Quantity – Number of products purchased
- Unit Price – Price per product
- Discount – Discount applied to the sale
- Tax – Tax applied to the sale
- Total Amount – Final price after calculations
- Payment Method – Method used for payment (Cash, Credit Card, etc.)
- Order Status – Status of the order (Completed, Processing, Cancelled)

Purpose

This table records all sales activities and connects customers, products, and locations.

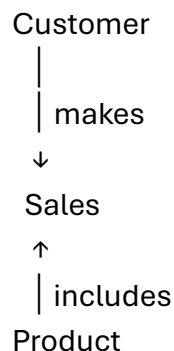
8. Relationships Between Tables

The tables in the system are connected using relationships.

Relationship Structure

- A **customer** can make many sales.
- A **product** can appear in many sales transactions.
- A **location** can have many sales.

This creates the following structure:





Location

This relationship ensures that all sales transactions are properly recorded and connected to the correct customer, product, and location.

9. OLTP Characteristics

This system follows the key characteristics of **Online Transaction Processing (OLTP)** systems.

9.1 Fast Processing

OLTP systems are designed to process transactions quickly. Sales transactions are recorded immediately when they occur.

9.2 Real-Time Data

The database is updated instantly whenever a transaction is completed. This ensures accurate and up-to-date information.

9.3 Data Integrity

The system prevents invalid data from being stored. For example, a sale cannot be recorded if the product does not exist.

9.4 Normalized Database

The database is structured to avoid duplication of data. Each piece of information is stored only once in the appropriate table.

9.5 ACID Properties

OLTP systems follow **ACID principles** to ensure reliable transactions.

- **Atomicity** – Transactions are completed fully or not at all
 - **Consistency** – The database remains accurate after transactions
 - **Isolation** – Transactions do not interfere with each other
 - **Durability** – Completed transactions are permanently stored
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10. Technical Components

The system uses several modern technologies to function properly.

10.1 Spring Boot

Spring Boot is a Java framework that simplifies the development of web applications and enterprise systems.

10.2 JPA and Hibernate

JPA (Java Persistence API) and Hibernate are used to map Java objects to database tables. This technique is known as **Object Relational Mapping (ORM)**.

10.3 Lombok

Lombok is a Java library that reduces repetitive code by automatically generating getters, setters, and constructors.

10.4 Maven

Maven is used to manage project dependencies and build the application.

10.5 H2 Database

The H2 database is a lightweight in-memory database used for testing and development. It resets when the application restarts.

11. System Features

The system allows users to perform several operations.

View Data

Users can view lists of:

- Customers
- Products
- Locations
- Sales records

Add New Records

Users can add new data such as:

- New customers
- New products
- New store locations
- New sales transactions

Database Console

Developers can access the H2 database console to view tables and run SQL queries.

Example access link:

<http://localhost:8080/h2-console>

12. Example Transaction Flow

The following steps describe how the system processes a typical sale.

1. A customer visits the store.
2. The system searches for the customer in the database.
3. The customer selects a product.
4. The system checks product availability.
5. The system calculates the total price including tax and discounts.
6. The payment method is selected.
7. The transaction is recorded in the Sales table.
8. The product stock quantity is reduced.
9. The transaction is completed successfully.

This entire process happens within seconds.

13. Learning Outcomes

This project demonstrates several important concepts in database and system development.

These include:

- Database design and normalization
 - Relationship modeling between tables
 - Development of enterprise applications using Spring Boot
 - Implementation of OLTP systems
 - Object Relational Mapping (ORM)
 - RESTful API development
 - Performance optimization using indexing
 - Data integrity and constraint management
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14. Conclusion

The OLTP Sales Management System provides a practical example of how businesses manage operational data using a database system. It demonstrates how real-time transactions can be processed efficiently using modern technologies such as Spring Boot and relational databases.

The system successfully manages customers, products, locations, and sales transactions while maintaining data integrity and fast performance. This project helps illustrate the principles of OLTP systems and highlights their importance in modern business applications.